



05506026

Digital Image Processing

Witchaya Towongpaichayont
Computer Science, KMITL

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EDGE DETECTION

Image Segmentation

- ▶ Recapping that in the previous slides, we deal with low-level processing where the inputs are images, and the outputs are images
- ▶ In the next couple of weeks, we will move on to mid-level processing where the inputs are images, and the outputs are image attributes

Image Segmentation

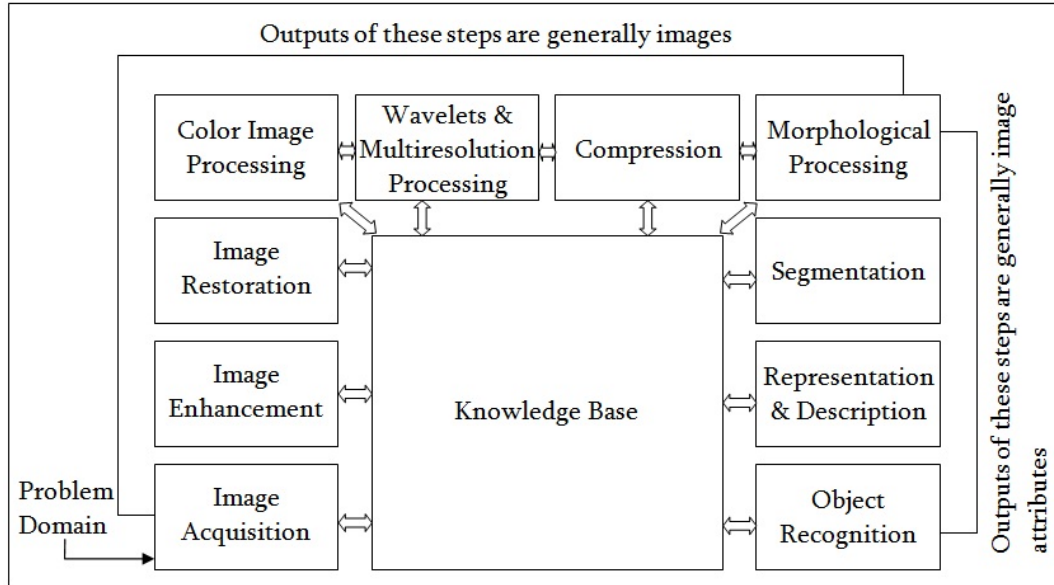


Image Segmentation

- ▶ Most of algorithms of basic image segmentation are based on one of two basic properties of image intensity values
 - ▶ Discontinuity
 - ▶ e.g. Edges, points, lines
 - ▶ Similarity
 - ▶ e.g. Thresholding, region growing, region splitting, region merging

Fundamentals of Image Segmentation

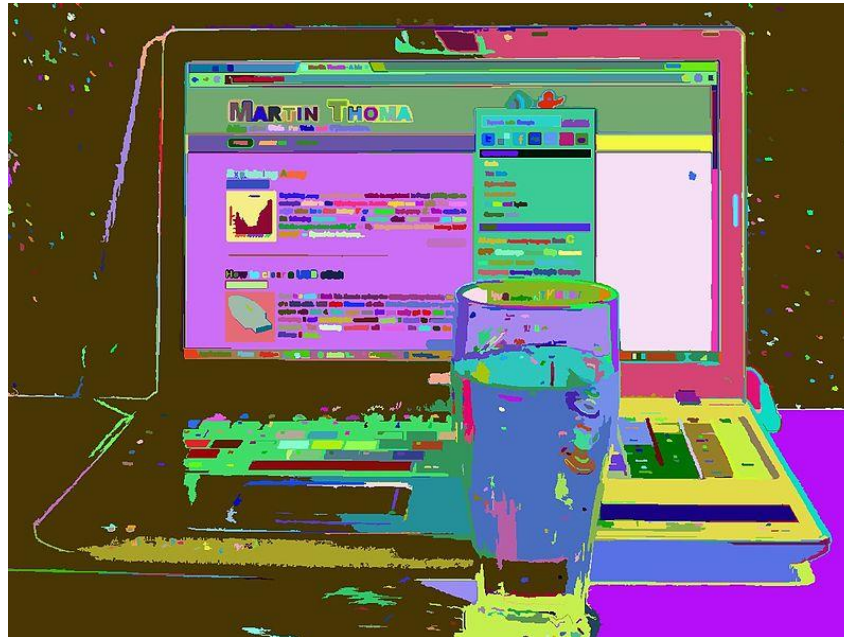


Image from: <https://commons.wikimedia.org/wiki/File:Image-segmentation-example-segmented-fe.jpg>

Fundamentals of Image Segmentation

- Image segmentation is a process that partitions the entire image, R , into n subregions, $R_1, R_2, \dots, R_n$, such that

a. $\bigcup_{i=1}^n R_i = R.$

b. R_i is a connected set, for $i = 0, 1, 2, \dots, n.$

c. $R_i \cap R_j = \emptyset$ for all i and $j, i \neq j.$

d. $Q(R_i) = \text{TRUE}$ for $i = 0, 1, 2, \dots, n.$

e. $Q(R_i \cup R_j) = \text{FALSE}$ for any adjacent regions R_i and $R_j.$

where $Q(R_k)$ is a logical predicate defined over the points in set $R_k,$



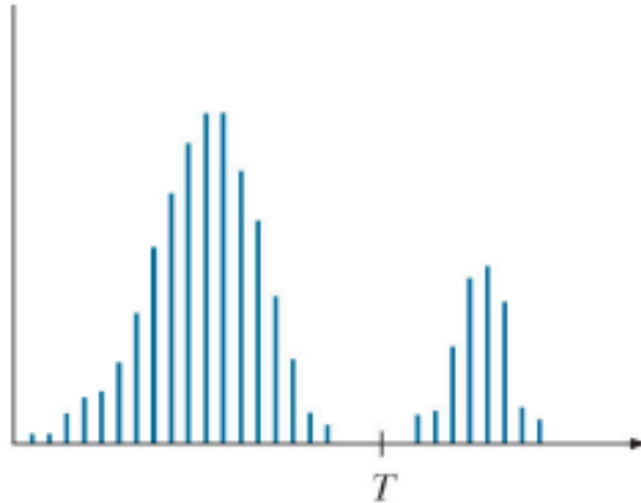
THRESHOLDING

Thresholding

- ▶ Thresholding is a non-contextual segmentation process to distinguish the regions of interest in an image from the rest.
- ▶ A Threshold value T is used to separate intensities of image into 2 groups

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > T \\ 0 & \text{if } f(x, y) \leq T \end{cases}$$

Thresholding



Thresholding

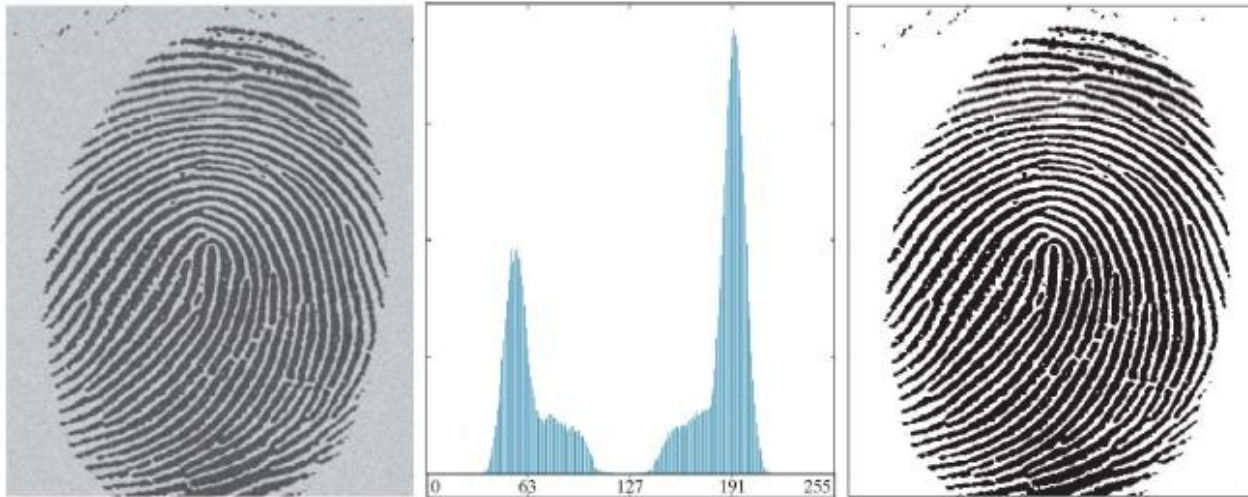
- ▶ If T is a constant, used over an entire image, it is called GLOBAL THRESHOLDING
- ▶ If T changes over time and space, it is called VARIABLE THRESHOLDING
- ▶ If T is used over a region, it is called LOCAL or REGIONAL THRESHOLDING

Basic Global Thresholding

Follow the iterative algorithm as follow

1. Select an initial estimate for the global threshold, T .
2. Segment the image using. This will produce two groups of pixels: G_1 consisting of pixels with intensity values $> T$ and G_2 , consisting of pixels with values $\leq T$
3. Compute the average (mean) intensity values m_1 and m_2 for the pixels in G_1 and G_2 respectively.
4. Compute a new threshold value midway between m_1 and m_2
$$T = \frac{1}{2}(m_1 + m_2)$$
5. Repeat Steps 2 through 4 until the difference between values of T in successive iterations is smaller than a predefined value, ΔT

Basic Global Thresholding



Global Thresholding using Otsu's Method

- ▶ Basic thresholding has been viewed as a statistical-decision theory problem which involves how to minimise the average error in two or more groups (as seen in the previous algorithm)
- ▶ However, this approach must involve the forms of PDF which are normally not in any recognisable shapes.
- ▶ One of popular approaches is to treat PDF form as some uniform functions such as Gaussian. But the process can be complex and not suitable for real-time applications

Global Thresholding using Otsu's Method

- ▶ Otsu's Method takes different approach. It uses the measure called BETWEEN-CLASS VARIANCE which is used in statistical discriminant analysis.
- ▶ The approach of Otsu's Method is to pick the Threshold values where they can best separate each class.

Global Thresholding using Otsu's Method

Follow the algorithm as follow

1. Compute the normalized histogram of the input image.

2. Compute the cumulative sums. $P_1(k) = \sum_{i=0}^k p_i$

3. Compute the average (mean) $m(k) = \sum_{i=0}^k ip_i$

4. Compute the global mean. $m_G = \sum_{i=0}^{L-1} ip_i$

5. Compute the between-class variance term $\sigma_B^2(k) = \frac{[m_G P_1(k) - m(k)]^2}{P_1(k)[1 - P_1(k)]}$

Global Thresholding using Otsu's Method

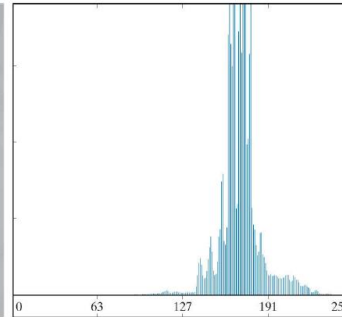
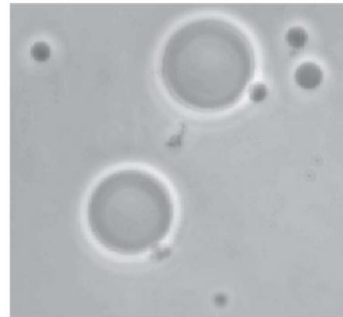
6. Obtain the Otsu threshold k^* as the value of k for which $\sigma_B^2(k)$ is maximum. If the maximum is not unique, k^* is the average of k which each creates the maximum.
7. Then, the input image $f(x, y)$ is segmented with

$$g(x, y) = \begin{cases} 1 & \text{if } f(x, y) > k^* \\ 0 & \text{if } f(x, y) \leq k^* \end{cases}$$

Global Thresholding using Otsu's Method



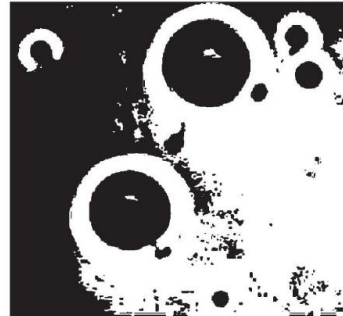
Original image



Histogram



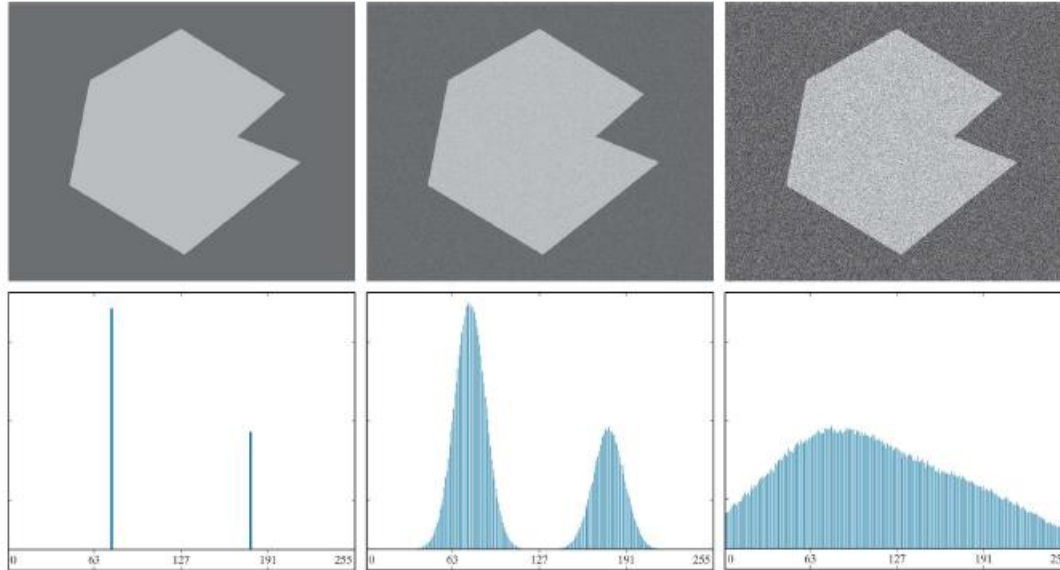
**Segmented using
basic global algorithm**



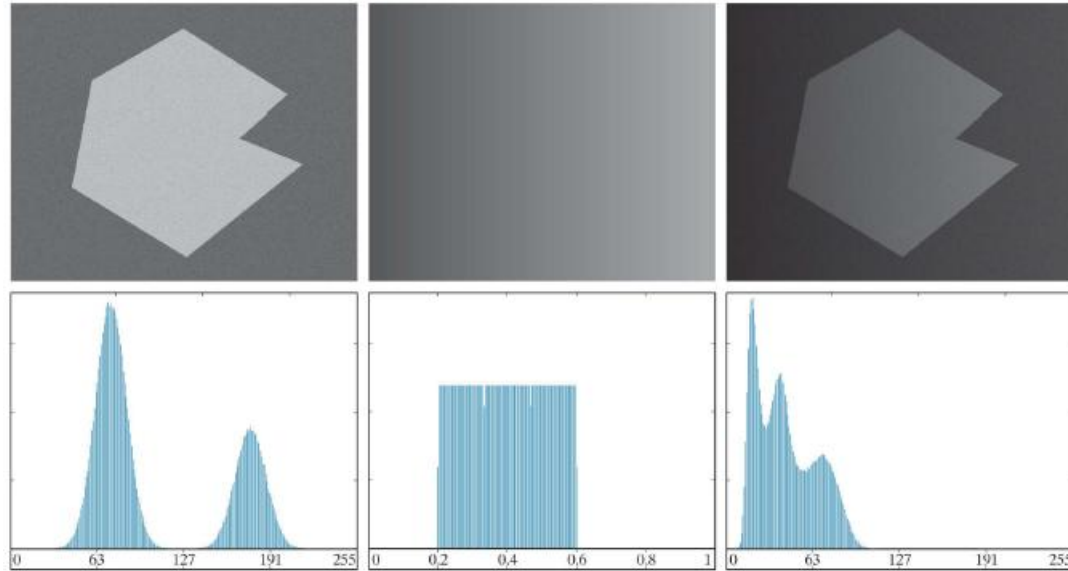
**Segmented using
Otsu's method**



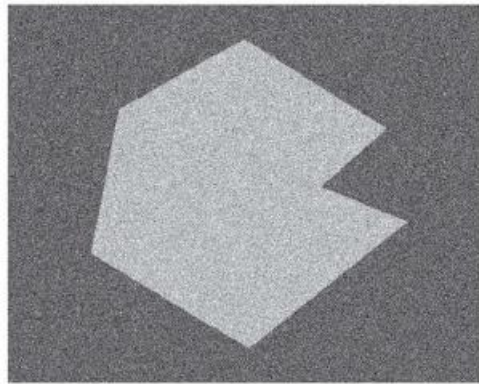
Noise in Thresholding



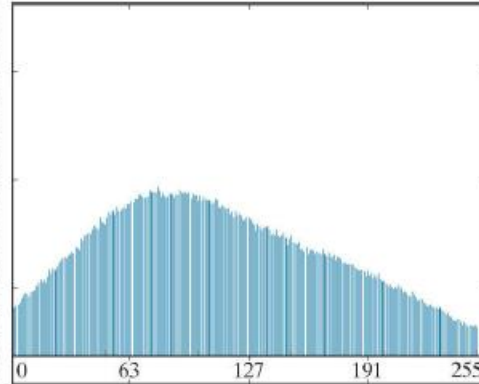
Illumination in Thresholding



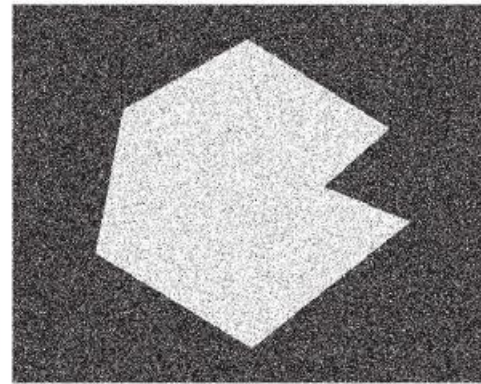
Improve Global Thresholding with Image Smoothing



Noisy image

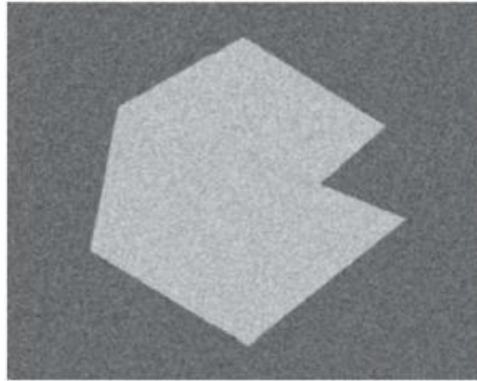


Histogram

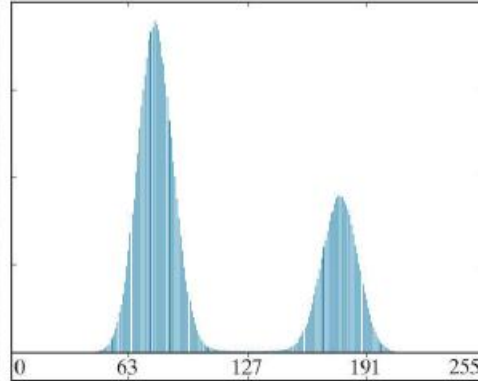


Result after using Otsu's method

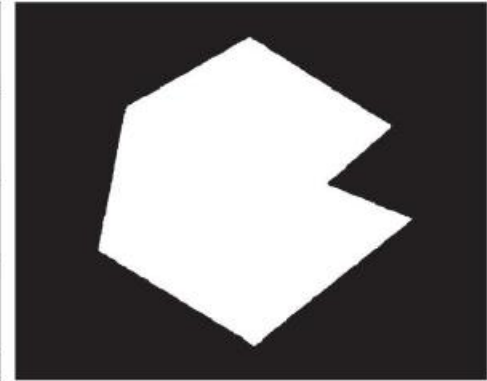
Improve Global Thresholding with Image Smoothing



**Smoothed with
5x5 averaging
kernel**



Histogram

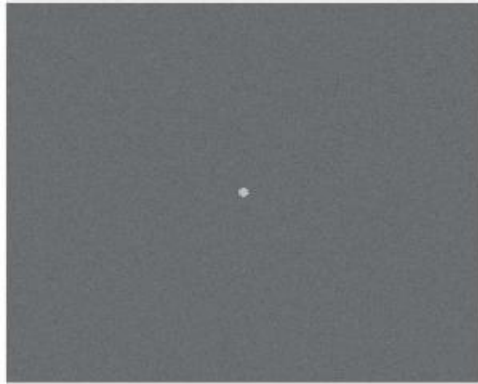


**Result after using Otsu's
method**

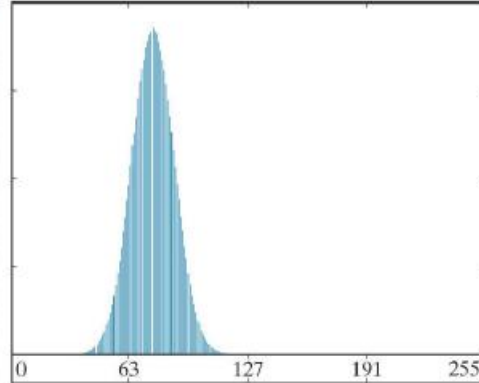
Improve Global Thresholding with Edges

- ▶ If the foreground size is extremely small respect to the background, the Thresholding might fail even we smooth the image
- ▶ This cannot be seen on the histogram

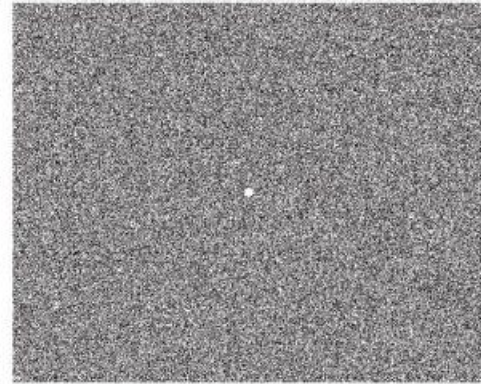
Improve Global Thresholding with Edges



Noisy image

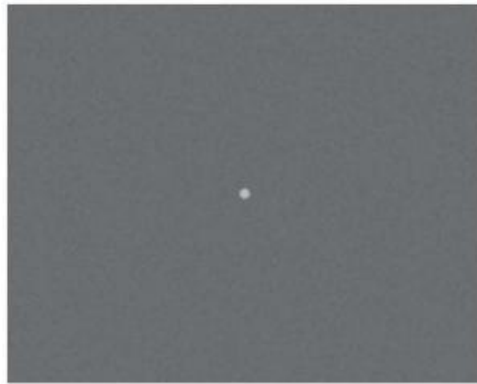


Histogram

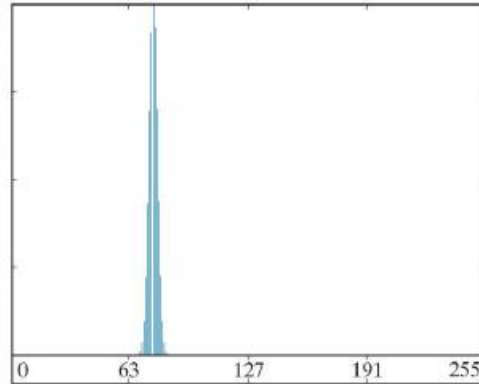


Result after using Otsu's method

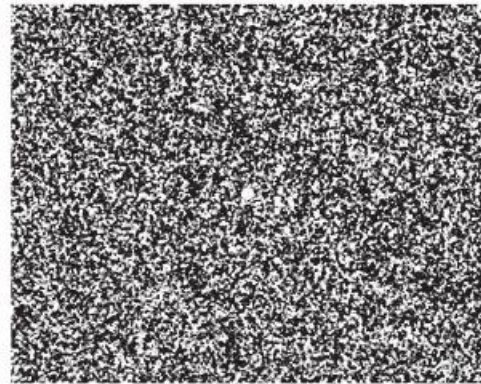
Improve Global Thresholding with Edges



**Smoothed with
5x5 averaging
kernel**



Histogram



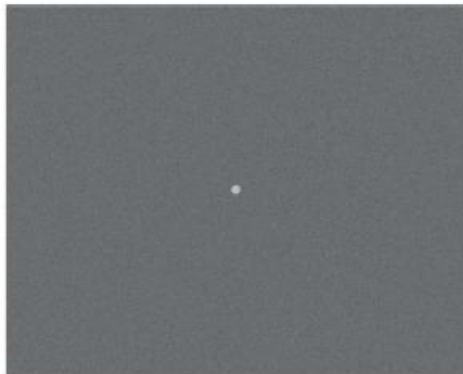
**Result after using Otsu's
method**

Improve Global Thresholding with Edges

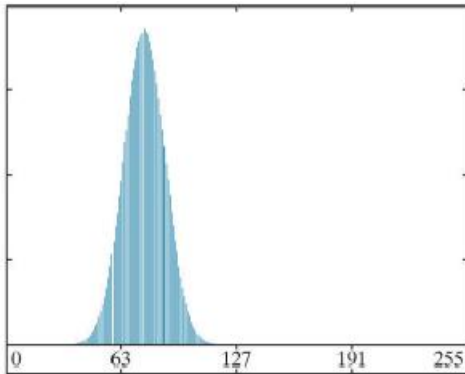
We can follow this algorithm so segment the image

1. Compute an edge image as either the magnitude of the gradient, or absolute value of the Laplacian, of $f(x, y)$ using any methods
2. Specify a threshold value, T .
3. Threshold the image from Step 1 using T from Step 2 to produce a binary image. This image is used as a mask image.
4. Compute a histogram using only the pixels in $f(x, y)$ that correspond to the locations of the 1-valued pixels in Step 3
5. Use the histogram from Step 4 to segment $f(x, y)$ globally using, for example, Otsu's method.

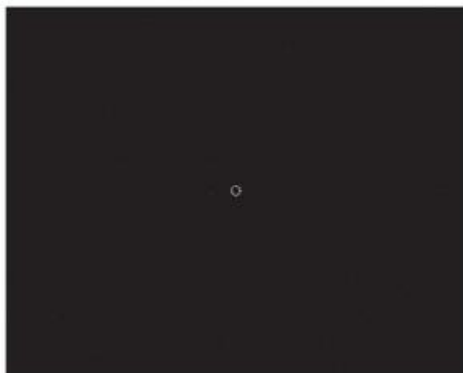
Noisy image



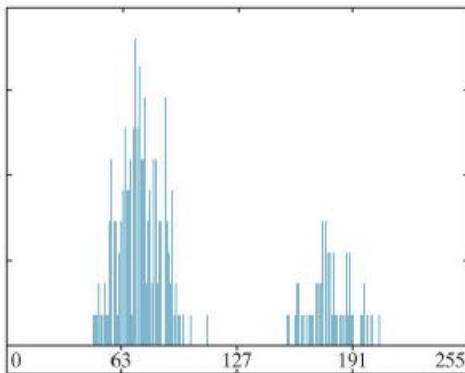
Histogram



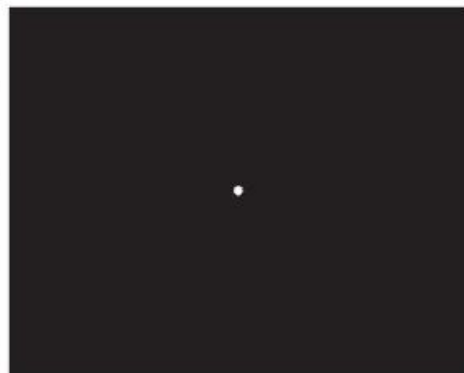
Mask image with the gradient magnitude image thresholded at 99.7 percentile



Product after masked

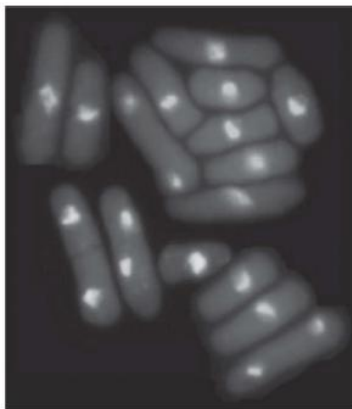


Histogram of masked image

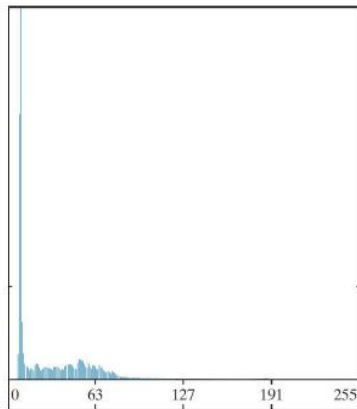


Result using Otsu's method

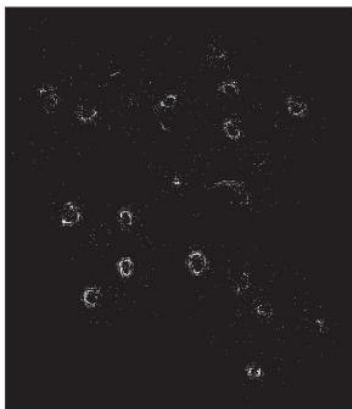
Original image



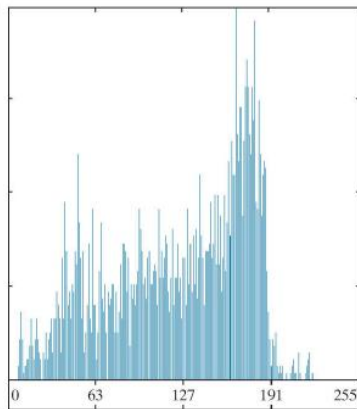
Histogram



**Result using
Otsu's method**



**Masked with
absolute Laplacian**



**Histogram of
masked image**



**Result using
Otsu's method**

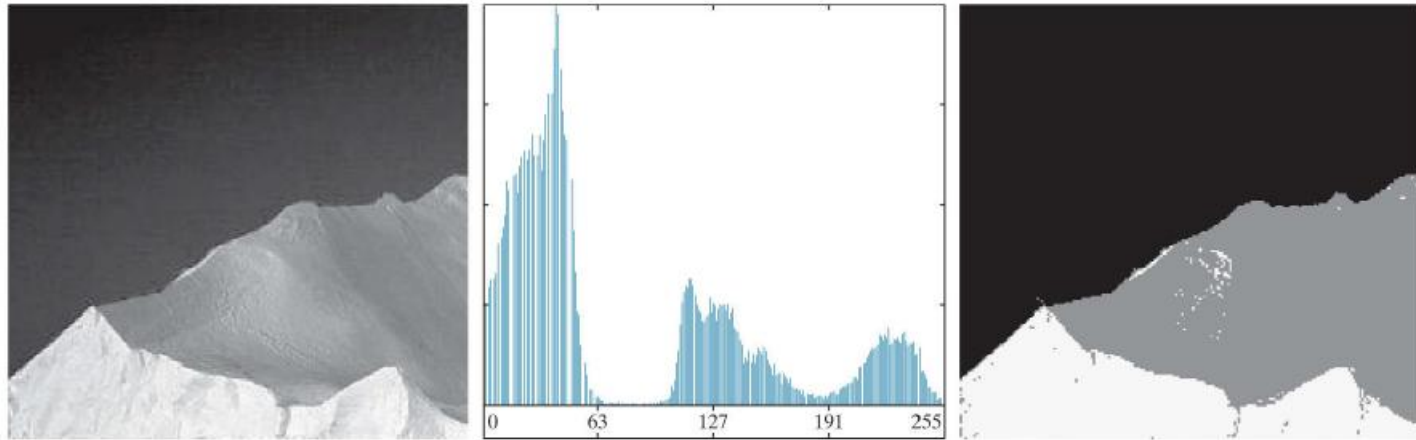


**Result using Laplacian
at 99.5 percentile and
Otsu's method**



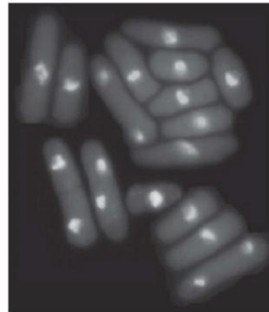
**Result using Laplacian
at 53.9 percentile and
Otsu's method**

Multiple Thresholding

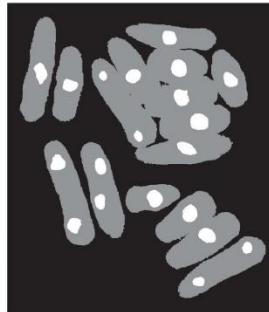


Variable Thresholding based on Local Image Properties

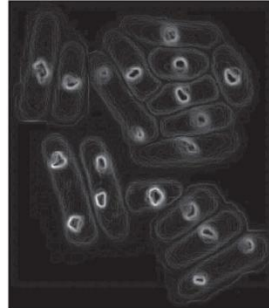
Original image



**Segmented using
dual thresholding**



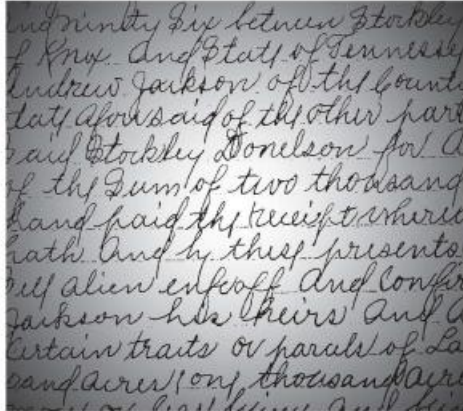
**3x3 local standard
deviation**



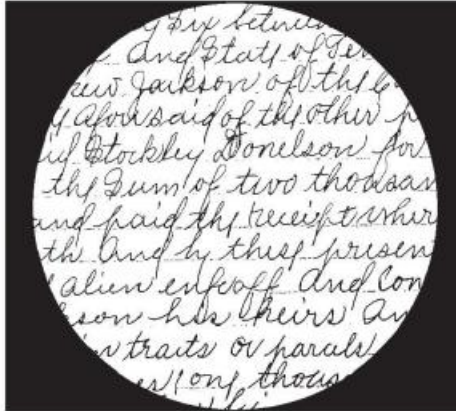
**Local thresholding
(2) – (3) and use
Global mean as
Threshold value**



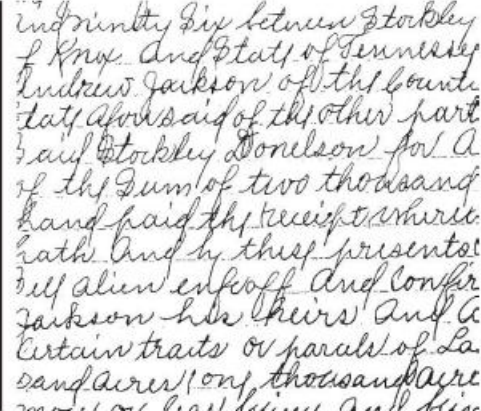
Variable Thresholding based on Moving Averages



Original image

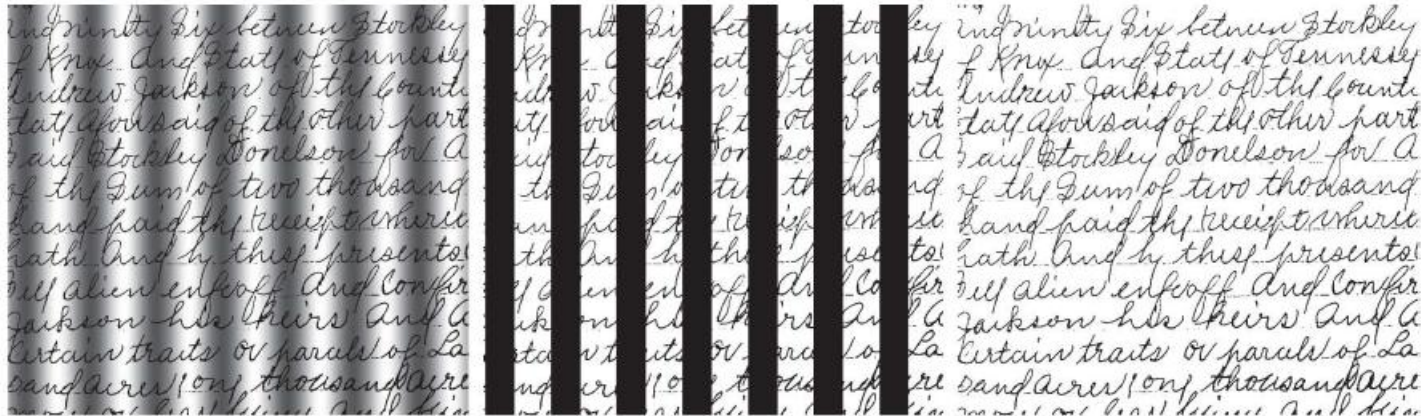


**Result of
Global Thresholding
using Otsu's method**



**Result of
Local Thresholding
using Moving Averages**

Variable Thresholding based on Moving Averages



Original image

**Result of
Global Thresholding
using Otsu's method**

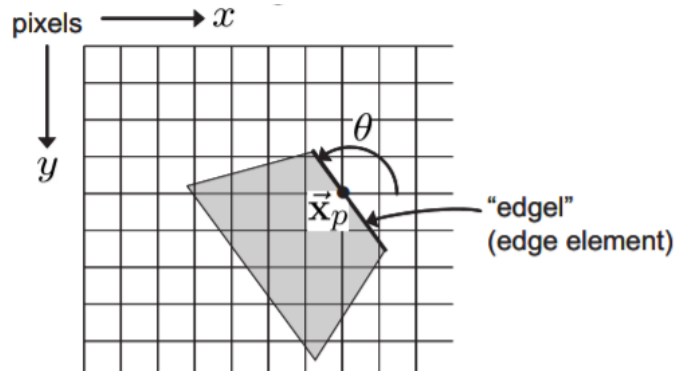
**Result of
Local Thresholding
using Moving Averages**



EDGE DETECTION

Point, Line, and Edge Detection

- ▶ Point, line, and edge mean that there are sharp and local changes in intensity, we call these together as EDGE PIXELS, sometimes called EDGELS (edge elements)
- ▶ Edges, or edge segments are sets of connected edge pixels

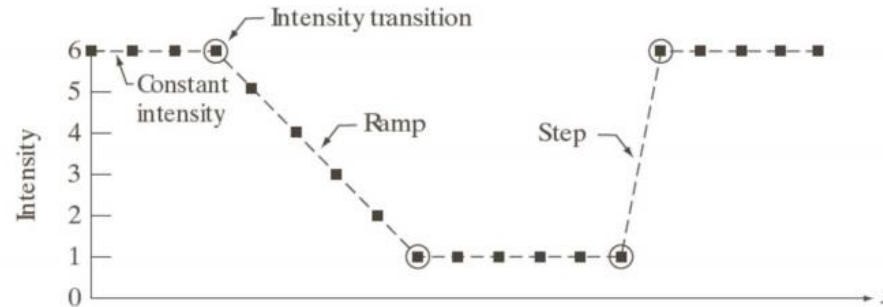


Point, Line, and Edge Detection

- ▶ Point is a foreground pixel surrounded by background pixels, or vice versa
- ▶ Line is a thin edge segment which the intensity of the background on either side of the line is either much higher or much lower than the intensity of the line pixels

Recapping image parts and derivatives

- ▶ We consider image in 3 parts
 - ▶ Flat segment (constant intensity)
 - ▶ Step
 - ▶ Ramp



Recapping image parts and derivatives

- ▶ First derivative

$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

x	
-1	1

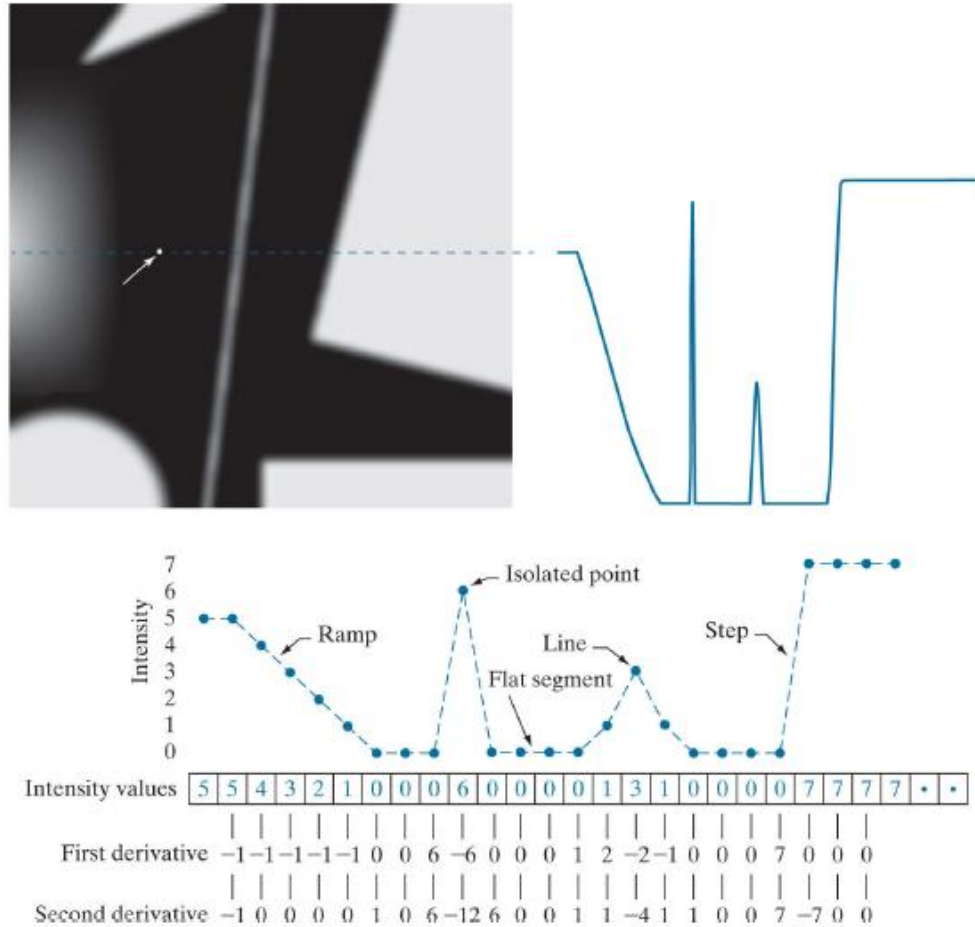
- ▶ Second derivative

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

x		
1	-2	1

Recapping image parts and derivatives

- ▶ First derivative: -
 - ▶ In the area of constant intensity, the value must be ZERO
 - ▶ At the onset of an intensity step or ramp, the value must be NONZERO
 - ▶ Along the ramp, the value must be NONZERO
- ▶ Second derivative: -
 - ▶ In the area of constant intensity, the value must be ZERO
 - ▶ At the onset AND at the end of an intensity step or ramp, the value must be NONZERO
 - ▶ Along the ramp WITH CONSTANT SLOPE, the value must be ZERO



Detection of Isolated Points

- ▶ Based on the previous section, point detection should be based on the second derivative

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

- ▶ When compute using the second-order finite differences, the equation will be

$$\nabla^2 f(x, y) = f(x + 1, y) + f(x - 1, y) + f(x, y + 1) + f(x, y - 1) - 4f(x, y)$$

Detection of Isolated Points

- Laplacian: -
$$g(x, y) = \begin{cases} 1 & \text{if } |Z(x, y)| > T \\ 0 & \text{otherwise} \end{cases}$$

1	1	1
1	-8	1
1	1	1

Detection of Isolated Points

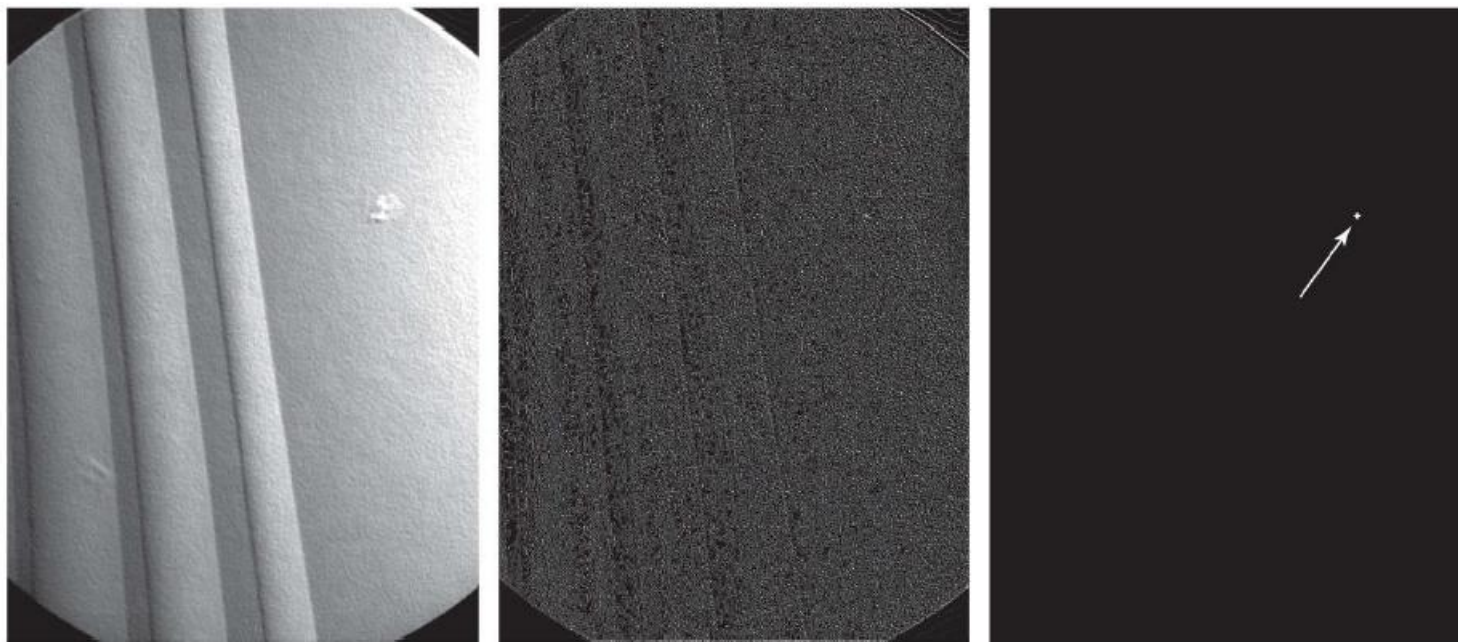
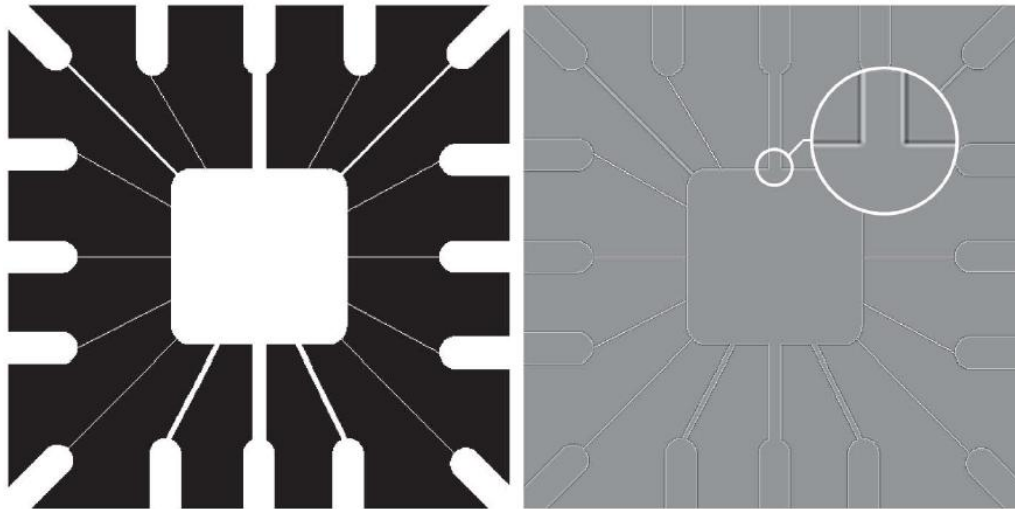


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.

Line Detection

- ▶ Line detection can also be executed using Laplacian kernel (second derivative). However, as second derivative can produce negative values, which cause double-line effect

Line Detection

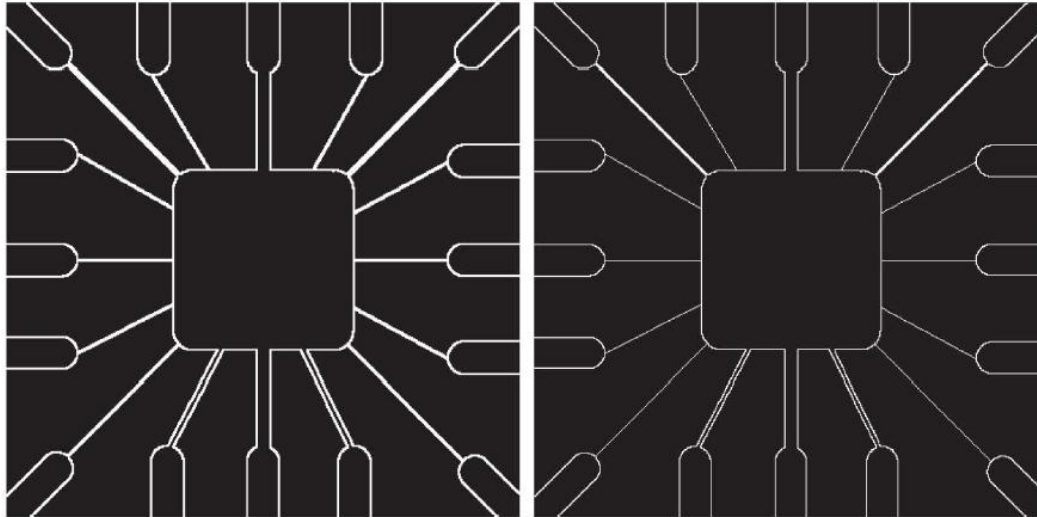


Original image

Laplacian image

Scaled by mid gray = 0, darker gray = negative values, lighter gray = positive values

Line Detection



**Absolute values
(double thickness of the lines)**

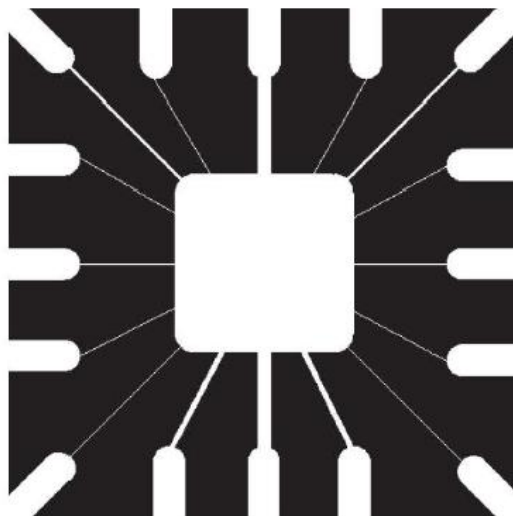
**Only positive values
(better approach)**

Line Detection

- ▶ Laplacian kernel normally detect edges which independent of directions. However, line detections are often interested in a specified direction. Therefore, kernels can be altered as follow.

-1	-1	-1	2	-1	-1	-1	2	-1	-1	-1	2
2	2	2	-1	2	-1	-1	2	-1	-1	2	-1
-1	-1	-1	-1	-1	2	-1	2	-1	2	-1	-1
Horizontal			+45°			Vertical			-45°		

Line Detection



Original image

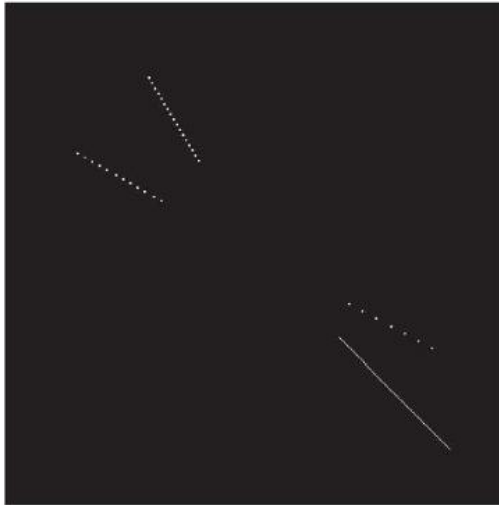


+45° filtered

Line Detection



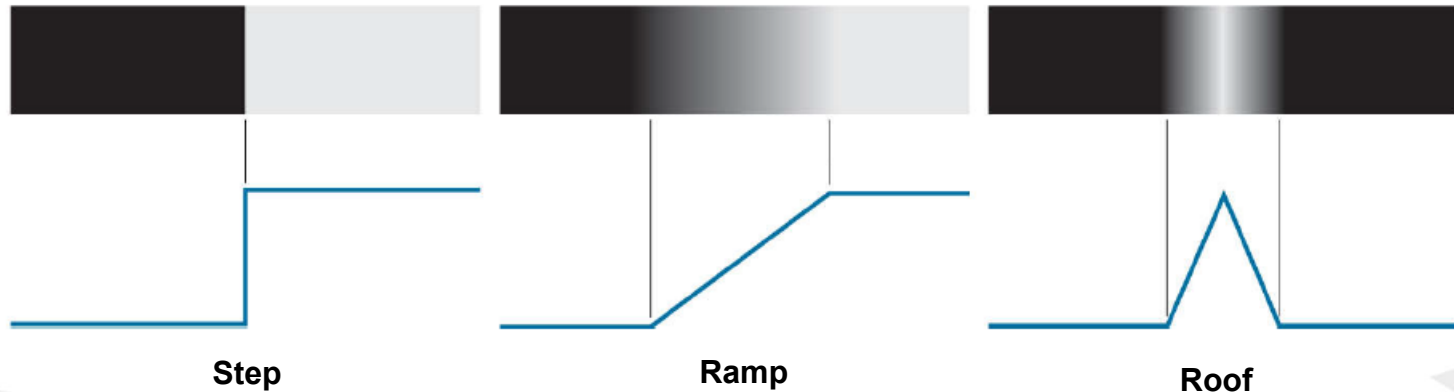
Threshold using mid value



Threshold using highest value

Edge Models

- ▶ Edges are the parts that their local intensities change sharply. There are three types of edge which ideally represented as follow



Edge Models

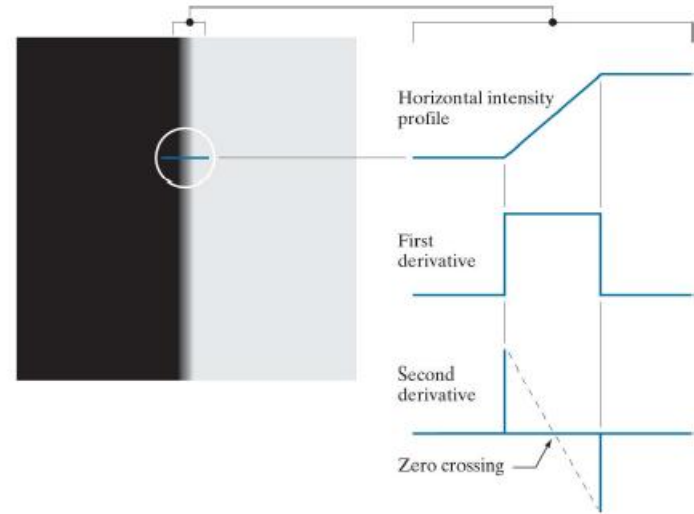
On an ideal ramp,

First derivative determines

- The presence of an edge

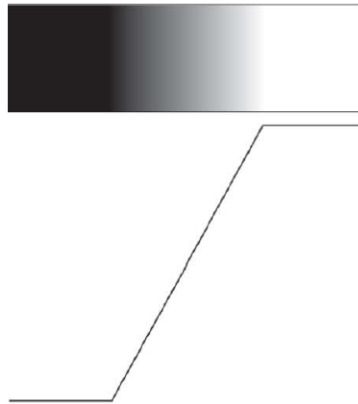
Second derivative determines

- Whether an edge lies on dark or light side
- Where the centres of thick edges are

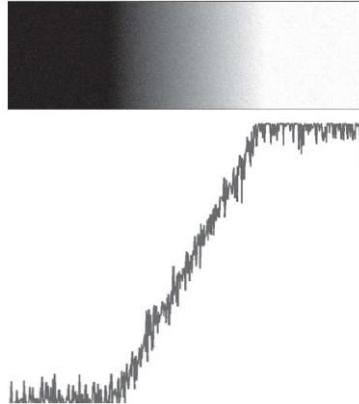


Edge Models

- ▶ However, in real practice, the edges can be blurred and noisy as follow



Ideal ramp



Noisy ramp

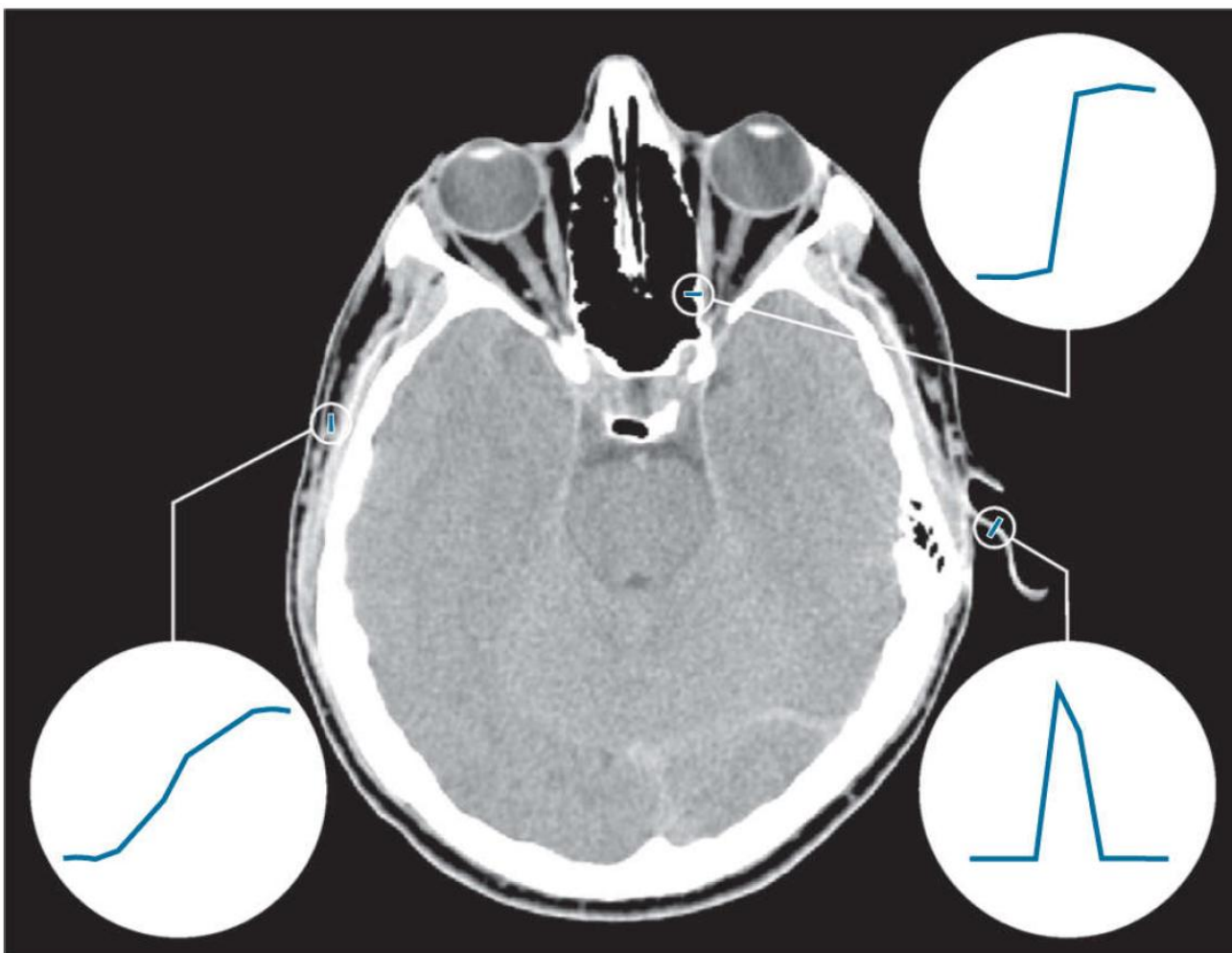
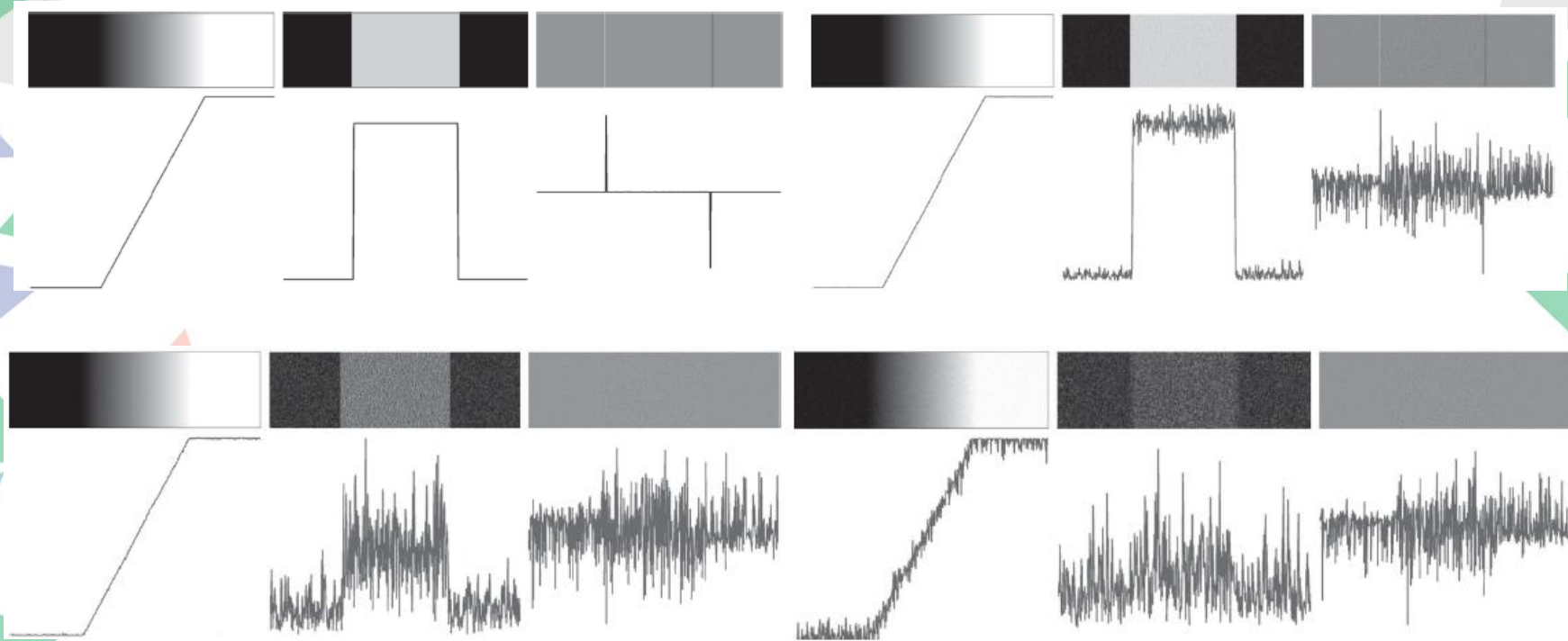


Image from: Gonzalez, R., & Woods, R. (2014). Digital image processing (3rd ed.). New Delhi: Dorling Kindersley.



Basic Edge Detection

- ▶ Three typical steps of edge detection
 - ▶ Image smoothing / noise reduction
 - ▶ Detection of edge point candidates
 - ▶ Edge localisation (select the points that are really edge points)

Original image



**Smoothed using
5x5 averaging
kernel**



**Edge detection
using Sobel's**



**Edge detection
using Sobel's
on smoothed
image**



Second derivative kernels

- ▶ Used for detecting the change of intensities or change in ramp
- ▶ More sensitive with noises
 - ▶ Laplacian

1	1	1
1	-8	1
1	1	1

First derivative kernels

- ▶ Used for detecting the direction of edge

- ▶ Sobel

-1	-2	-1	-1	0	1
0	0	0	-2	0	2
1	2	1	-1	0	1

- ▶ Prewitt

-1	-1	-1	-1	0	1
0	0	0	-1	0	1
1	1	1	-1	0	1

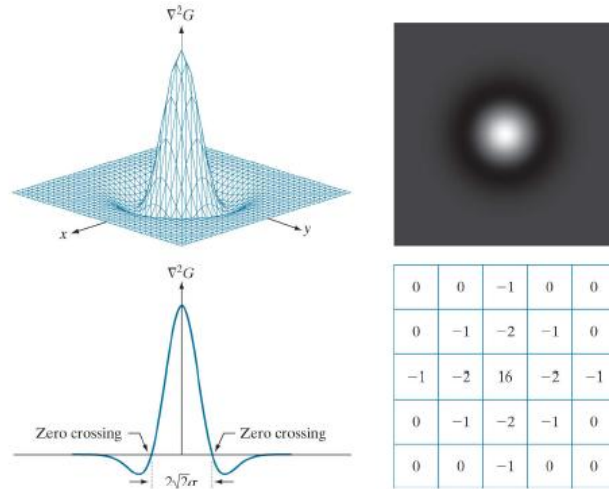
First derivative kernels

- ▶ Used for detecting the direction of edge
 - ▶ Kirsch compass

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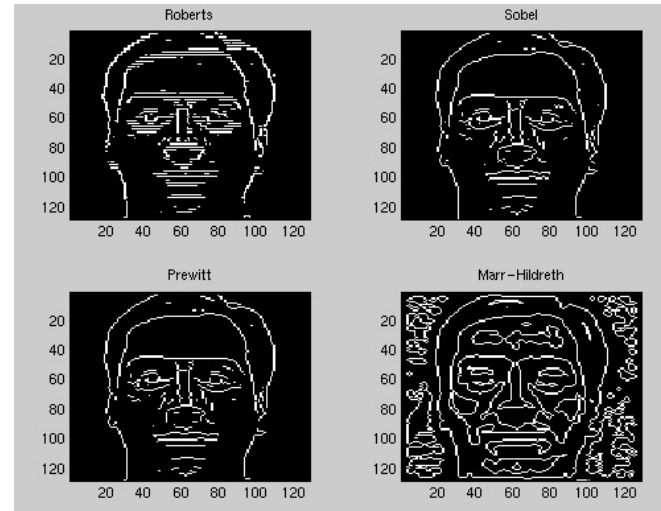
First derivative kernels

- ▶ Used for detecting the direction of edge
 - ▶ Marr-Hildreth



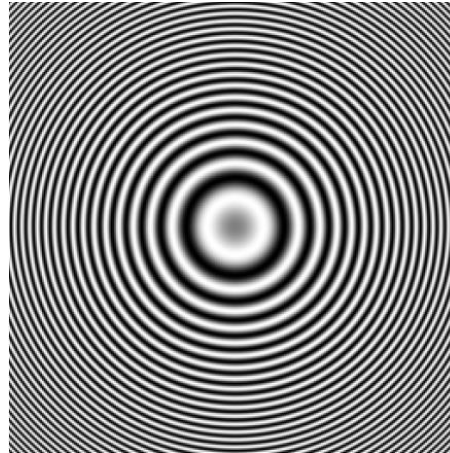
First derivative kernels

- ▶ The differences between each kernels are
 - ▶ Sensitivity to noises
 - ▶ Behaviours at corners
 - ▶ Complexities of image
 - ▶ Texture-based image
 - ▶ Computation time
 - ▶ Etc.



First derivative kernels

- ▶ One image that is used a lot to compare the differences is Jähne test patterns



The Canny Edge Detector

- ▶ Canny Edge Algorithm is more complex, but the performance is known that superior in general. Canny's approach is based on three basic objectives
 - ▶ Low error rate
 - ▶ Edge points are located as close as possible to the centre of true edges
 - ▶ Return only single edge point

The Canny Edge Detector

- ▶ The process of Canny Edge Detection algorithm has 4 steps
 1. Apply Gaussian filter to smooth the image

$$G(x, y) = e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

2. Find the intensity gradient

$$M_s(x, y) = \|\nabla f_s(x, y)\| = \sqrt{g_x^2(x, y) + g_y^2(x, y)}$$

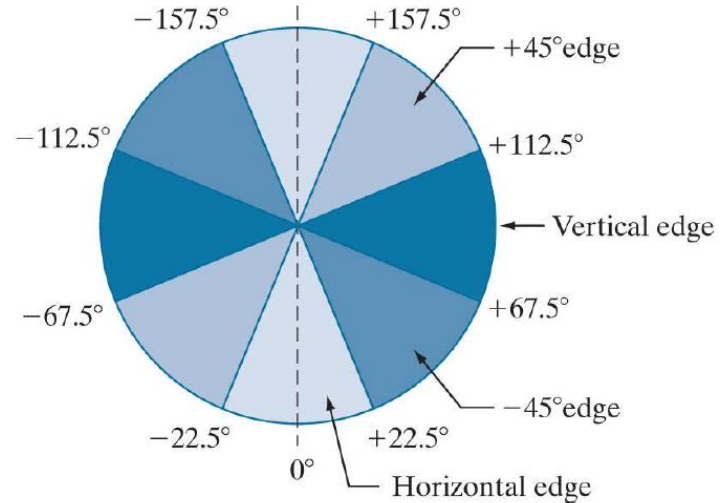
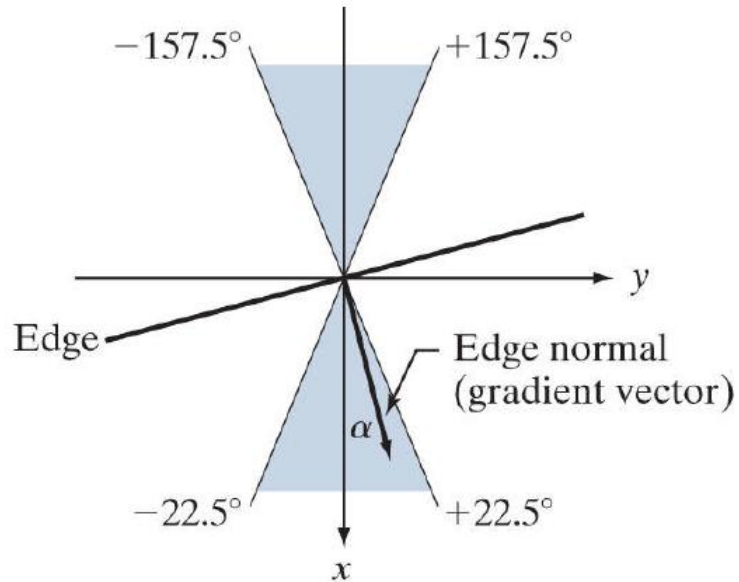
$$\alpha(x, y) = \tan^{-1} \left[\frac{g_y(x, y)}{g_x(x, y)} \right]$$

The Canny Edge Detector

3. Apply non-maximum suppression to thin the ridges from step 2. The process is to specify a number of possible discrete orientations of edge normal vector.

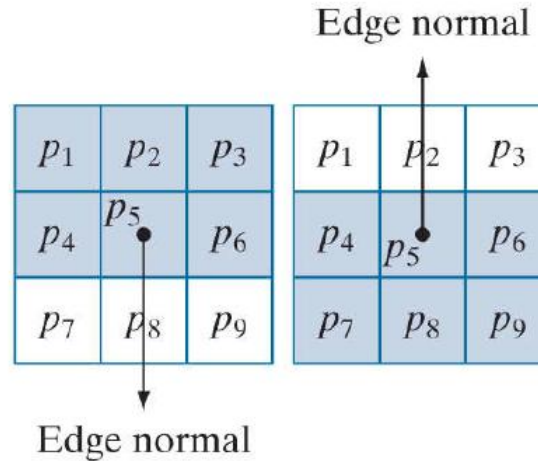
For example, 3 x 3 region can define four orientations for the edge: -horizontal, vertical, +45°, and -45°. Therefore, we can determine the orientation using ranges of the degree of the edge

The Canny Edge Detector



The Canny Edge Detector

Note that every edge has two possible orientations



The Canny Edge Detector

Non – maximum suppression steps are

1. Find the direction d_k that is closest to $\alpha(x, y)$.
2. Let K denote the value of $\|\nabla f_s\|$ at (x, y) . If K is less than the value of $\|\nabla f_s\|$ at one or both of the neighbors of point (x, y) along d_k , let $g_N(x, y) = 0$ (suppression); otherwise, let $g_N(x, y) = K$.

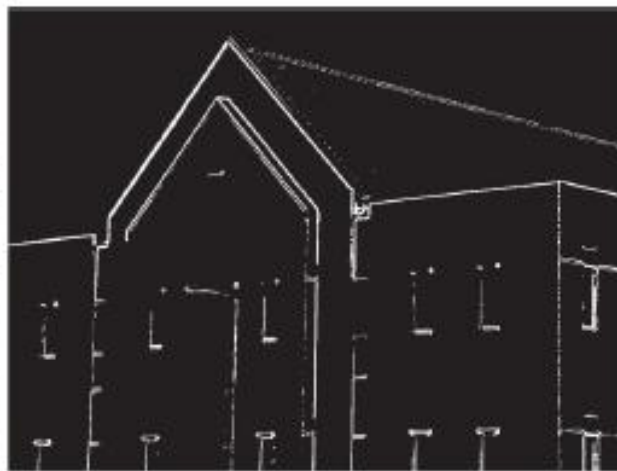
The Canny Edge Detector

4. Apply double threshold, hysteresis thresholding

Original image



**Thresholded
gradient of the
smoothed image**



Marr-Hildreth



Canny algorithm

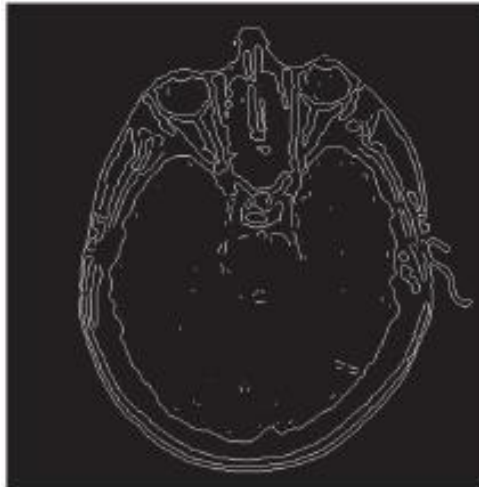


Original image

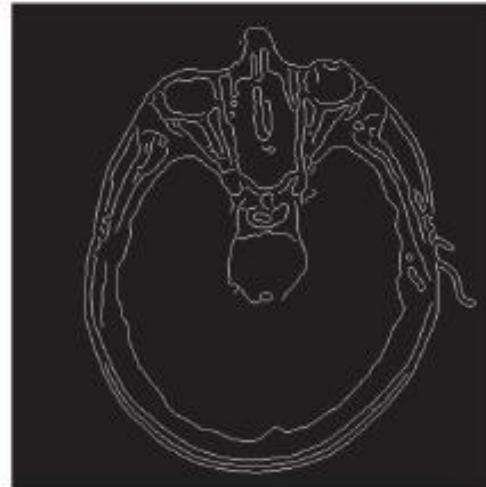


**Thresholded
gradient of the
smoothed
image**

Marr-Hildreth



**Canny
algorithm**





End of Week 9